

Static yaw misalignment from high-resolution SCADA: the label-free (T0) method, step by step

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Abstract

This note derives the label-free (tier T0) method behind `Results_33_T0_0.csv` (PPP_WTG17) and `Results_33_T0_final.csv` (SSS_WTG06). It is the T0 part of the complete derivation note of the project: every operation performed on the data is described in the order in which it is performed, with the exact thresholds that the code uses. Section 1 fixes the notation, §2 covers reconstruction and cleaning, §3 the sensor model, §4 the neighbour residual and the states, §5 the submission model, §6 the scoring rule and §7 the assumptions. The method fits no parameter to labels; §5.5 states exactly where labels were looked at. The code that implements the method is in `src/yaw/` and `scripts/` of this repository.

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1 Notation

1.1 Conventions

All directions are compass bearings in degrees, clockwise from north. A difference of two bearings is always wrapped to $[-180^\circ, 180^\circ)$,

$$\text{wrap}(x) = ((x + 180) \bmod 360) - 180. \quad (1)$$

Time has three resolutions. The raw stream is sampled at roughly 12s; a raw sample is indexed by t . Minute aggregates are indexed by m (1-minute bins) or m_{10} (10-minute bins). Calendar days are indexed by d . Turbines are indexed by i (the target) and j (a neighbour). A contiguous run of days with constant behaviour is a *state*, indexed by ℓ .

For a set of bearings $x_1, \dots, x_n$ the *circular mean* and the *mean resultant length* are

$$\bar{x} = \text{atan2}\left(\frac{1}{n} \sum_i \sin x_i, \frac{1}{n} \sum_i \cos x_i\right), \quad \bar{R} = \left| \frac{1}{n} \sum_i e^{ix_i} \right| \in [0, 1], \quad (2)$$

where $\bar{R} = 1$ means all bearings are equal and $\bar{R} \rightarrow 0$ means they are spread around the circle. The *circular median* is the ordinary median of the wrapped deviations from the circular mean,

$$\text{med}^\circ(x) = \bar{x} + \text{med}_i \text{wrap}(x_i - \bar{x}). \quad (3)$$

It inherits the robustness of the median and does not depend on where the $\pm 180^\circ$ cut sits. med without subscript is the ordinary median.

1.2 Symbols

Table 1 lists every symbol used in the note. Physical quantities that carry the undisclosed scale factor of the data set (power, wind speed, rotor and generator speed, pitch) are only ever used in ratios, differences or threshold comparisons, never as absolute physical values.

<i>Reported SCADA channels (per raw sample t)</i>	
P	active power (scaled units; rated $P_{\text{rated}} \approx 4800$)
U	nacelle-anemometer wind speed (scaled)
Ω, Ω_g	rotor and generator speed (scaled)
φ	blade pitch angle (degrees)
N_r	reported nacelle heading, field <code>NacDir</code>
W_r	reported wind direction, field <code>WindDir</code>
v	reported vane angle, derived: $v = \text{wrap}(W_r - N_r)$
<i>Physical (unobserved) quantities</i>	
W_t	true free-stream wind direction at the rotor
N_t	true nacelle heading (rotor axis)
γ	instantaneous yaw error $\gamma = \text{wrap}(W_t - N_t)$, positive when the wind comes from clockwise of the rotor axis
θ	static yaw misalignment $\theta = \mathbb{E}[\gamma]$ over a day or a state; the label
δ_n, δ_v	mounting offsets of the nacelle encoder and of the wind vane
g	gain of the vane ($g \leq 1$); the implementation uses $g = 1$
s	vane set-point of the yaw controller
<i>Consensus and states</i>	
D_{ij}	heading difference of a pair, $D_{ij} = \text{wrap}(N_r^i - N_r^j)$
J	size of a level shift in a pair series
$m_{ij}(\cdot)$	sector median of D_{ij} in the neighbour's wind-direction sector
κ	unknown constant of the consensus (§4.1)
$r_i(d)$	daily consensus heading residual of turbine i
$\tilde{r}_i(d)$	its 7-day rolling median
$\bar{r}_\ell$	level of state ℓ (median of $\tilde{r}$ over the state)
$\lambda, n_{\min}$	change-point penalty and minimum state length of PELT
<i>Submission</i>	
L	level prior of a turbine (vane set-point)
α	anchor of an encoder-free segment
x	centred state level $x = \bar{r}_\ell - \alpha$

Table 1: Symbols. Hats denote estimates, bars denote means or levels.

2 Reconstruction, cleaning and aggregation

2.1 The change-based stream

The parquet files hold one row per turbine and timestamp with seven channels ($P, U, \Omega, \Omega_g, \varphi, N_r, W_r$). The stream is *change-based*: a channel is written only when its value changed, and a null means “unchanged since the previous row”, not “missing”. Per channel the fraction of written rows is 0.5–0.8, except N_r , which is written on only $\approx 6\%$ of rows because the nacelle moves only during yaw manoeuvres.

Reconstruction rule. Rows are sorted by (turbine, t). Before filling, the indicator

$$u_t = \mathbf{1}[N_r \text{ written at } t] \quad (4)$$

is kept: it marks a *yaw controller action* and is lost after filling. Then every channel is forward-filled per turbine (`ffill`), so that each row carries the last known value of every channel. Nothing is interpolated; a value holds until the next write.

The vane. The vane angle is not shipped. In this SCADA the wind direction field is computed by the controller as heading plus vane, so

$$v_t = \text{wrap}(W_r - N_r)_t \quad (5)$$

returns the vane reading exactly (checked on the raw pairs). Consequently W_r carries whatever offset the nacelle encoder carries, and only v is encoder-free. This single fact shapes the whole pipeline (§3).

Asynchronous writes (a trap). Because N_r and W_r are written independently, at a row where N_r was just updated the filled W_r still holds the value computed at the *previous* W_r write. Equation (5) then reads $v^{\text{old}} - \Delta$ (with Δ the heading change) until W_r refreshes, whatever the physical vane does. Nothing at the 12-second scale may therefore use v across a heading update; see §3.3.

2.2 Aggregation with circular statistics

The filled stream is binned to 1-minute and 10-minute bins. For a bin containing raw samples $t \in m$:

- linear channels $(P, U, \Omega, \Omega_g, \varphi, v)$: the mean $\bar{P}_m$, etc.; in addition the within-bin standard deviation of P , U and v , and the median v_m^{med} of the vane;
- directions (N_r, W_r) : the circular mean (2) and the resultant length $\bar{R}_m$; a naive mean of 350° and 10° would be 180° , and several turbines spend much time near north;
- counts: number of raw rows n_m , number of controller actions $\sum_{t \in m} u_t$, and the operating fraction $\omega_m = \frac{1}{n_m} \sum_{t \in m} \mathbf{1}[P_t > P_{\min}]$.

Per turbine this gives $\approx 1.0 \times 10^6$ minutes from $4.7\text{--}5.9 \times 10^6$ raw rows over the 727–731 days of record.

2.3 Operating regime

The undisclosed scale factor forbids physical thresholds; the thresholds are therefore set from the data. The 99.9th percentile of P defines $P_{\text{rated}} = 4800$. A minute is

$$\text{operating} \iff \bar{P}_m > P_{\min} = 100 \quad (\approx 2\% \text{ of rated}). \quad (6)$$

This is the only use of power in the T0 method.

2.4 Daily table and sensor-health flags

Per turbine and day the minutes are summarised: number of minutes, rows and controller actions, operating minutes, and, over the *operating* minutes only, the daily median vane

$$v_d^{\text{med}} = \text{med}_{m \in d, \text{operating}} v_m^{\text{med}}, \quad (7)$$

its inter-quartile range, the mean within-minute vane standard deviation, the median pitch and rotor speed, and the circular mean heading. Three flags mark days that no estimator may use:

$$\begin{aligned} \text{frozen vane} &: \text{mean within-minute vane std} < 0.05^\circ \wedge \text{operating minutes} > 60, \\ \text{no yaw} &: \sum_d u_t = 0 \wedge \text{operating minutes} > 60, \\ \text{sparse} &: \text{raw rows} < 1000. \end{aligned}$$

A day is *healthy* when none of the three is set. The flags catch two real faults: PPP_WTG13 reports $v = -34.0^\circ$ with zero variance and no controller action on 8–16 Feb 2024 and $v = +39.1^\circ$ on 9–12 Oct 2024 (11 and 7 days); no other turbine has a frozen vane. Sparse days are 1–8 per turbine.

2.5 Labels and label-derived states

The label θ_d^* is given per row of the training turbines and is constant within a day; the daily series is its first value per day (195, 312 and 195 labelled days on PPP_WTG12/13/14, January to November 2023). The organisers score only days outside a transition window around each label step, but do not publish the window rule; it is reconstructed as follows. The daily label series of each turbine is segmented into piecewise-constant states with the PELT algorithm (§4.4) under an L_2 cost with penalty $\lambda = 20$ and minimum length $n_{\min} = 7$ days. The state level is the median label within the state. A day is a *transition* when it lies within 7 calendar days of a state boundary (gaps count as days) or when its label deviates by more than 0.75° from its state level; a day is *scored* when it is labelled and not a transition. This gives 2, 6 and 2 states and 180, 217 and 180 scored days. All local scoring (§6) uses this table.

2.6 Layout and neighbours

The layout files give coordinates with an unknown common scale, so distances are only used as ratios and bearings are exact. The neighbours of turbine i are the same-site turbines sorted by distance.

2.7 Pair offsets

For every same-site pair (i, j) and every 10-minute bin in which both turbines operate more than 90 % of the bin ($\omega_{m_{10}} > 0.9$ for both), the heading difference $D_{ij} = \text{wrap}(N_r^i - N_r^j)$ and the wind-direction difference $\text{wrap}(W_r^i - W_r^j)$ are formed, and their daily medians and inter-quartile ranges are stored (65 pairs). This table is diagnostic (Figure 1); the consensus of §4 recomputes the pair series with a circular median.

2.8 What the cleaning revealed

Three facts from this step drive the design:

1. **The vane set-point is not zero.** On every PPP turbine the daily median vane in operation is $-2.7 \dots -3.7^\circ$; on SSS it is -0.8 or -3.1° , and SSS_WTG06 wanders between 0 and -4° over weeks. The controller nulls the vane at $\approx -3^\circ$, not at 0.
2. **Encoders lose north.** PPP_WTG12’s heading jumps by $\approx 138^\circ$ against every neighbour from August 2023 while its label does not move; on 10–17 % of days a turbine of the WTG12/13/14 cluster sits $100\text{--}180^\circ$ away from all neighbours for several days while producing normally. These are encoder-frame events, not flow.
3. **Labels and vane can move in opposite directions.** On PPP_WTG13 the vane median moved from ≈ -0.5 to $\approx -3^\circ$ on 14–18 January 2023 while the label went from -11.5 to -8.1° . This becomes the sign check of §5.4.

3 The sensor model

3.1 True and reported quantities

At any instant the unobserved quantities are the true wind direction W_t , the true nacelle heading N_t and the yaw error $\gamma = \text{wrap}(W_t - N_t)$. The stream reports

$$N_r = N_t + \delta_n, \tag{8}$$

$$v = g\gamma + \delta_v, \tag{9}$$

$$W_r = N_r + v. \tag{10}$$

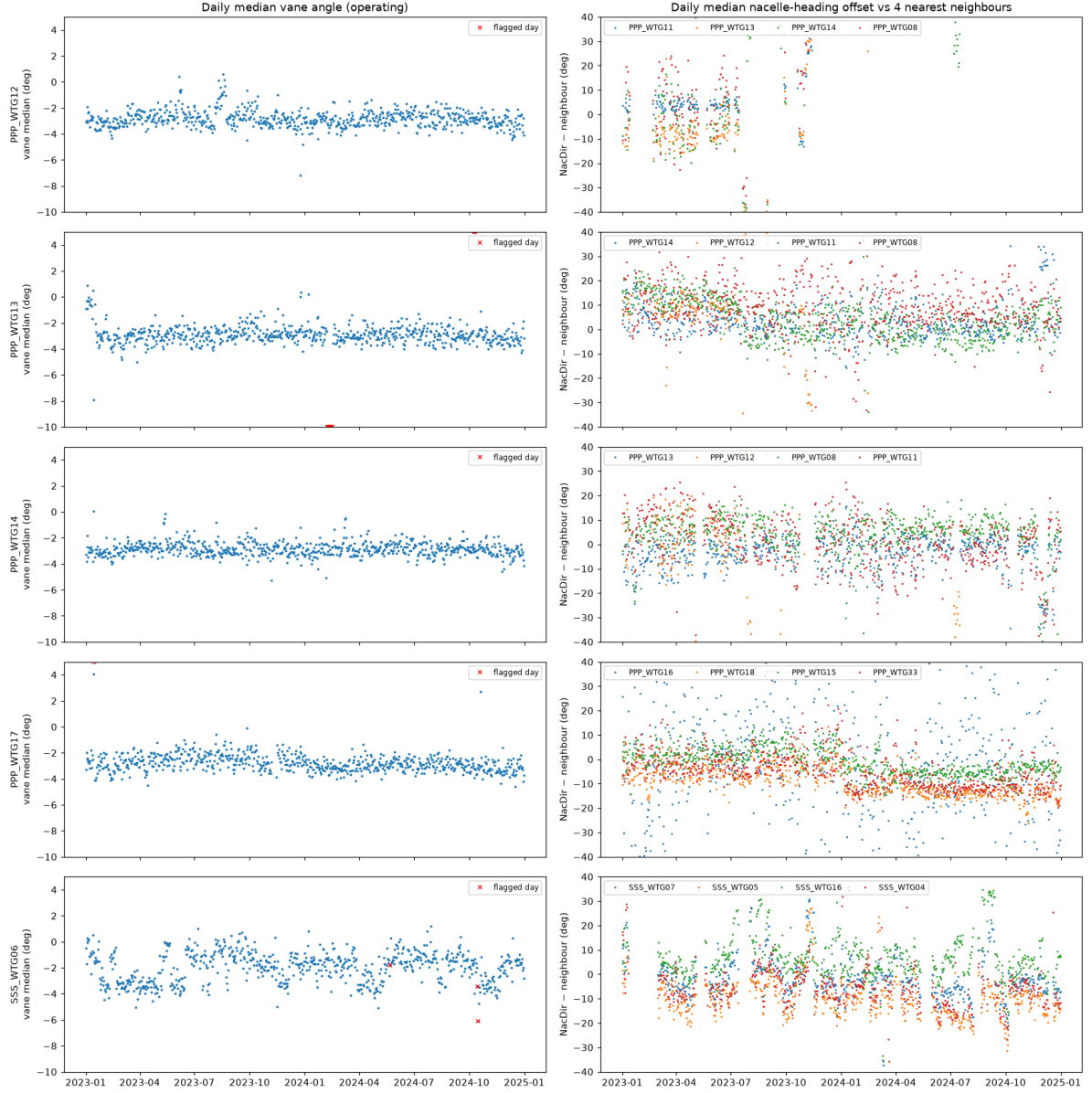


Figure 1: Left: daily median vane angle in operation, v_d^{med} . Right: daily median heading offset D_{ij} to the four nearest neighbours, before de-stepping and cleaning.

Equation (8) says the encoder has no north reference: δ_n is an arbitrary constant that changes at every re-reference. Equation (9) says the vane measures the yaw error with a mounting offset δ_v and a gain $g \leq 1$; the gain models the well-documented damping of the inflow angle behind a rotating rotor (wake swirl, nacelle body).¹ Equation (10) is not an assumption but the encoding of the data set, §2.

3.2 The controller and the definition of the label

The yaw controller filters v over minutes and rotates the nacelle whenever the filtered reading leaves a deadband (measured occupancy $\approx \pm 8^\circ$) around its set-point s . In steady state the controller therefore enforces $\mathbb{E}[v] = s$, and from (9)

$$\theta \equiv \mathbb{E}[\gamma] = \frac{s - \delta_v}{g}. \quad (11)$$

θ is the *static yaw misalignment*: the persistent yaw error the controller cannot see because it sits inside its own measurement. The WindFit reference system measures it with an independent sensor and fits the power-optimal heading over several days, which is why its daily series is smoothed and drifts for about a week before each step, and why days near a step are not scored. The label is negative on all three training turbines; the sign check of §5.4 establishes that the label sign coincides with γ as defined here.

Two consequences shape everything that follows.

1. **The vane cannot see θ .** Averaged over any window the vane reads s whatever θ is. Measured: $v_d^{\text{med}} \approx -3^\circ$ on every PPP turbine irrespective of the label (correlation with the label -0.33). Information about θ must come from the *response* of the turbine to γ (power) or from an external direction reference (neighbours). The T0 method uses the neighbours.
2. **θ changes in three ways** that the sensors do not treat alike: a change of the set-point s , of the vane offset δ_v , or of the gain g . All three move θ by (11); only the first two move the neighbour residual of §4; and an encoder re-reference (δ_n) moves the residual without moving θ at all.

3.3 Why the vane gain cannot be measured here

If the vane responded instantaneously with gain g , a nacelle rotation Δ during steady wind would produce a vane jump of $-g\Delta$ ((9) with $\gamma \rightarrow \gamma - \Delta$), and the thousands of manoeuvres would give g directly. On this data the derived vane jumps by *exactly* $-\Delta$ at every heading update. The reason is the asynchronous encoding described in §2: at a heading update W_r still holds $N_r^{\text{old}} + v^{\text{old}}$, so (5) reads $v^{\text{old}} - \Delta$ until W_r refreshes, whatever the vane does. The apparent gain is one by construction, and no quantity derived from the two direction channels at the 12-second scale can recover g . Therefore $g = 1$ is used throughout.

4 Encoder frames, consensus residual and states

4.1 Sensor algebra of the residual

Suppose a consensus estimate $c(t)$ of the true wind direction is available from the neighbours, of the form $c = W_t + \kappa$ with κ a constant that depends on the neighbours' own offsets. From (8) and $\gamma = W_t - N_t$,

$$\text{wrap}(N_r - c) = N_t + \delta_n - W_t - \kappa = -\gamma + \delta_n - \kappa. \quad (12)$$

¹Everything below is written for general g ; the implementation uses $g = 1$ because §3.3 shows g cannot be measured from this data.

Averaged over a day, $\mathbb{E}[\gamma] = \theta$, so the daily residual obeys

$$\boxed{r(d) = -\theta(d) + \delta_n(d) - \kappa} \implies \Delta r = -\Delta\theta \quad \text{whenever } \Delta\delta_n = 0. \quad (13)$$

Proposition 1. *The residual identifies changes of the static misalignment in physical degrees with slope exactly -1 , independent of the vane (g, δ_v) and of the aerodynamics; its level is not identifiable, because δ_n and κ are unknown constants.*

This is the “consensus anchor” of the leading team. Using W_r instead of N_r gives $W_r - c = \delta_n + \delta_v + (g - 1)\gamma - \kappa$, a constant for $g = 1$ and a damped version of (13) for $g < 1$; the implementation uses N_r . The residual is blind to changes of θ that occur through g (the wind side of γ does not move), but such changes are also invisible to the label mechanism to first order.

The consensus c is never formed explicitly. Instead the residual is built pair by pair, §4.3, which is equivalent because every pair difference $D_{ij} = \text{wrap}(N_r^i - N_r^j)$ contains $-\gamma^i + \gamma^j + \delta_n^i - \delta_n^j$ and the neighbour’s terms are constant over a day up to noise.

4.2 Pair series, excursions and re-references

δ_n is piecewise constant with jumps. A jump of δ_n^i moves D_{ij} against *all* neighbours j by the same amount on the same day; the detection rule uses exactly this signature. Two kinds of jump occur and must be treated differently: a *re-reference*, after which δ_n stays at its new value, and an *excursion*, in which the heading reads of the order of 100° off for one to six weeks and then returns to the old frame. Excursions are by far the more frequent.

Reference set. For each turbine i the $n_{\text{nb}} = 4$ nearest same-site turbines are its references, excluding PPP_WTG12 and PPP_WTG16, whose headings are too unstable to serve as references (they still receive their own residual from the remaining neighbours).

Daily pair series. From the 10-minute aggregates, for bins where both turbines operate more than 90 % of the time,

$$D_{ij}(d) = \text{med}_{m_{10} \in d}^\circ \text{wrap}(N_r^i - N_r^j)_{m_{10}}. \quad (14)$$

Level shifts. Each daily pair series is unwrapped (so that a crossing of $\pm 180^\circ$ is not read as a jump) and segmented with PELT under an L_1 cost, penalty $\lambda = 150$, minimum length $n_{\text{min}} = 10$ days (§4.4). The L_1 cost (median per segment) is chosen so that single outlying days do not create segments. Each boundary yields a candidate jump J , the wrapped difference of the medians of the adjacent segments.

Attribution rule. A candidate jump on date d in the series (i, j) is credited to the encoder of i when candidates with $|J| \geq 25^\circ$ of the same sign appear within ± 3 days in at least 60 % of i ’s pairs and in at least two of them. The credited jump is the median of the matching candidates and its date the earliest of them. Shifts below 25° cannot be attributed by majority: a 5° misalignment change and a 5° re-reference look identical in every pair. Fifty-nine steps on nine turbines were credited, none on PPP_WTG17 or its three nearest neighbours.

Why the credited jumps are not subtracted. The first version of the method subtracted every credited jump J_n from the heading from its date on. For an excursion this subtracts two independent estimates, one out and one back, each a difference of medians of whole PELT segments of which the excursion segment holds only 10–20 noisy days. They do not cancel: on SSS_WTG06 the pairs $(-92.2, +101.9)$, $(-119.2, +125.7)$, $(-111.0, +112.4)$ and $(-108.9, +117.3)$ leave $+9.6$, $+6.5$, $+1.5$ and $+8.4^\circ$ in r , and a one-week deviation in November 2023 was credited

as a single -30.3° step with no return. These remainders are as large as the misalignment signal of (13) and produced 13 states on SSS_WTG06 where the raw pair offsets show three. The validate turbine has no credited step and was unaffected, so the defect was invisible in the validate round.

Frame plan. The credited step dates $d_1 < d_2 < \dots$ of turbine i cut its record into intervals; the jump sizes J_n are not used further.

1. An interval shorter than 45 days is an *excursion*. Its days are dropped, together with ± 3 days around every step date. Nothing is subtracted.
2. For two consecutive long intervals A (before) and B (after) the shift is measured *locally*,

$$J_{AB} = \text{med}_j \text{ wrap } (\bar{D}_{ij}^B - \bar{D}_{ij}^A), \quad \bar{D}_{ij}^B = \text{med}_{d \in B_{40}}^\circ D_{ij}(d), \quad \bar{D}_{ij}^A = \text{med}_{d \in A_{40}}^\circ D_{ij}(d), \quad (15)$$

with B_{40} the first 40 days of B and A_{40} the last 40 days of A , on the raw daily pair series (14), with a margin of 3 days at the interval edges, at least 7 days on each side, and without the days within 25 days of a credited step of the neighbour j .

3. If $|J_{AB}| < 15^\circ$ the frame *came back*: no correction is applied, and whatever small difference remains between A and B is left in the data as signal.
4. If $|J_{AB}| \geq 15^\circ$ the shift is a *re-reference*: from the start of B on, both direction channels are corrected,

$$N_r^i(t) \leftarrow (N_r^i(t) - J_{AB}) \bmod 360, \quad \text{and likewise } W_r^i, \quad (16)$$

cumulatively over successive re-references, and the date is recorded.

W_r must be corrected too because it carries δ_n by (10); v is unaffected. The plan finds 37 excursions on eight turbines, 10 to 43 days long (117 days in seven excursions on SSS_WTG06, 123 on SSS_WTG04, 148 on PPP_WTG14), and a single re-reference: PPP_WTG12 on 2023-07-28, $J_{AB} = 147^\circ$. The rule assumes that the misalignment does not change by 15° or more across an excursion; a change below that survives, a change above it would be removed as a re-reference.

4.3 Building the residual

After the frame plan has been applied, three cleaning operations and one correction are applied to each pair before the residual is formed.

1. **Transient removal.** The daily pair series (14) is recomputed on the corrected headings and compared with its centred rolling median over 91 days with data (at least 7 present; gaps are skipped, so the window can span more calendar days). Days with $|\text{wrap}(D_{ij}(d) - \text{rolling median})| > 10^\circ$ are dropped from the pair. A rolling median reproduces a level shift of any size exactly but suppresses a pulse shorter than half its window, so this removes the excursions, of the target *or* of the neighbour, that are too small or too short to be credited in §4.2 (two of about 18° on SSS_WTG06), while a persistent step passes unchanged. The price is that a real state shorter than about 45 days which departs by more than 10° and returns is removed as well. Lowering the 25° attribution threshold to 12° instead was tried and did not catch these excursions.
2. **Sector correction.** Terrain steering and partial wakes make the direction difference between two sites a function of the wind direction. With σ_j the 30° sector of the neighbour's (de-stepped) wind direction, $\sigma_j = \lfloor W_r^j / 30^\circ \rfloor$, the sector median $m_{ij}(\sigma_j) = \text{med}^\circ \{D_{ij}(m_{10}) : \text{bins in sector } \sigma_j\}$ is subtracted bin by bin,

$$e_{ij}(m_{10}) = \text{wrap}(D_{ij}(m_{10}) - m_{ij}(\sigma_j(m_{10}))). \quad (17)$$

The neighbour’s wind direction is used as the sector variable because it is not affected by the target’s own yaw error.

3. **Daily pair residual.** $e_{ij}(d) = \text{med}_{m_{10} \in d} e_{ij}(m_{10})$ over days with at least 12 valid bins; its daily inter-quartile range is kept as a noise measure.
4. **Noisy references.** A pair whose median daily inter-quartile range exceeds twice the median of the turbine’s pairs is dropped.

The daily residual of turbine i is the median over the surviving pairs,

$$r_i(d) = \text{med}_j e_{ij}(d). \quad (18)$$

The double median (over bins, then over neighbours) makes (18) robust to one bad neighbour and to a few bad bins; the measured day-to-day noise is $\approx 2^\circ$ (median absolute deviation). The price is that changes below $\approx 1^\circ$, the two small label steps on WTG12 and WTG14, are under the noise floor. In terms of §4.1, the sector medians m_{ij} absorb δ_n^j , θ^j and the terrain term into the constant κ , and r_i is (13) up to that constant.

4.4 Change-point detection: PELT

Both the label states (§2.5), the encoder steps (§4.2) and the SCADA states (§4.5) use the same algorithm. Given a series $y_1, \dots, y_T$, a partition into contiguous segments $\mathcal{S}_1, \dots, \mathcal{S}_M$ is chosen to minimise

$$\sum_{\ell=1}^M C(\mathcal{S}_\ell) + \lambda(M-1), \quad |\mathcal{S}_\ell| \geq n_{\min}, \quad (19)$$

with the segment cost either

$$C_{L_2}(\mathcal{S}) = \sum_{d \in \mathcal{S}} (y_d - \bar{y}_\mathcal{S})^2 \quad \text{or} \quad C_{L_1}(\mathcal{S}) = \sum_{d \in \mathcal{S}} |y_d - \text{med}(y_\mathcal{S})|. \quad (20)$$

The pruned exact linear time algorithm (PELT) solves (19) exactly by dynamic programming over the last change point, discarding candidates that can never become optimal. The L_2 cost is the negative log-likelihood of a piecewise-constant Gaussian mean up to scale, so λ acts as a BIC-type penalty: a break is accepted when it lowers the residual sum of squares by more than λ . For a level jump J over n days on both sides, the reduction is $nJ^2/2$, so with $\lambda = 400$ a jump is detected when $nJ^2/2 \gtrsim 400$, e.g. $J \geq 2.4^\circ$ over 140 days or $J \geq 5^\circ$ over 30 days. The penalties were chosen on the training turbines: lower values split PPP_WTG17’s 2023 into up to 13 states and SSS_WTG06 into 50, noise rather than physics.

4.5 SCADA states

The residual is put on a complete daily calendar, smoothed with a centred 7-day rolling median (at least 3 days present) and linearly interpolated across gaps so that states are contiguous in calendar time,

$$\tilde{r}_i(d) = \text{med}\{r_i(d') : |d' - d| \leq 3\}. \quad (21)$$

PELT with the L_2 cost, $\lambda = 400$ and $n_{\min} = 14$ days is run on $\tilde{r}_i$. Each state ℓ receives the level

$$\bar{r}_\ell = \text{med}_{d \in \mathcal{S}_\ell} \tilde{r}_i(d). \quad (22)$$

The first day of each state is typed *encoder* when $|\bar{r}_\ell - \bar{r}_{\ell-1}| > 15^\circ$ (too large for a misalignment change) or when it lies within 10 days of a re-reference recorded by the frame plan of §4.2, and *candidate* otherwise; days within ± 7 of a boundary are flagged as transitions, mirroring the scoring rule of §2.5.

Measured on the training turbines against the label-derived states of §2.5: adjusted Rand index 0.77 / 0.31 / 0.00 on WTG12/13/14, limited by the sub-degree label steps that r cannot see (PPP_WTG14 receives a single state; its label step is 0.7°). On WTG13 the residual gives one boundary, on 2023-08-23, with $\Delta r = -5.1^\circ$ for a label step of $+3.65^\circ$ four weeks earlier. *A small re-reference and a misalignment change are indistinguishable in r* : r gives boundaries reliably and magnitudes only as a prior. On PPP_WTG17 the result is two states with the boundary at 2024-01-06 and $\Delta r = -7.74^\circ$, i.e. $\Delta\theta = +7.7^\circ$ by (13). On SSS_WTG06 it is six states, none with an encoder-typed boundary: boundaries on 2023-06-18 and 2023-09-25 with $\Delta r = +8.5^\circ$ and -7.9° , and three further states in 2024 that differ by 2 to 6° . The residual exists on 519 of the 731 days of this turbine; the first 63 days fall into one long excursion.

5 The submission model

5.1 Structure

For every turbine-day the prediction is

$$\hat{\theta}(d) = L - x(d), \quad x(d) = \bar{r}_{\ell(d)} - \alpha_{\text{seg}(\ell(d))}, \quad (23)$$

where $\ell(d)$ is the day’s SCADA state (§4.5), $\text{seg}(\ell)$ its *encoder-free segment* (a maximal run of states without an encoder-typed boundary, i.e. the cumulative count of encoder boundaries up to ℓ), α the segment anchor and L the level prior. The slope of -1 on x is (13); no parameter is fitted. The submitted `cluster` field is $\ell(d)$. Days without a SCADA state (gaps at the record edges) take the nearest state’s values; predictions are rounded to three decimals. The clusters are computed from the residual over all 731 days without any knowledge of the template or of which days are scored; the only calendar prior is $n_{\min} = 14$ days.

5.2 The level prior L

The residual gives changes only; a measured level would have to come from the turbine’s power response to γ . With the standard empirical law $P \propto \cos^k \gamma$, $k \approx 2$, the power varies by $\frac{k}{2}(8\pi/180)^2 \approx 2\%$ over the $\pm 8^\circ$ deadband of the yaw controller, against sector-, turbulence- and season-dependent power variations of the same order. Estimators of the peak location built on this law were evaluated in the full project; none survived as a level estimator and none is used here. The level is therefore a prior, the *vane set-point*: from (11) with $g = 1$ and the assumption $\delta_v = 0$, $\theta = s$, and s is measured as

$$L = \text{med}_d \text{ healthy } v_d^{\text{med}}, \quad (24)$$

the median over healthy days of the daily median vane in operation (7). On the PPP site this is -2.7 to -3.7° on every turbine, on SSS -0.8 to -3.3° . The prior is right when the vane is mounted true and the operator has chosen a non-zero set-point deliberately; it is wrong by $-\delta_v$ otherwise. On the training turbines it gives biases of -1.0 , $+2.3$ and -1.2° for the complete model: about a degree on two of three, and more than two degrees on WTG13, whose vane evidently carries its own offset.

5.3 The anchor α

The prior (24) is a statement about the turbine’s *typical* state, so the state levels must be centred before the slope is applied. Two choices were evaluated on the training turbines, both taken over the days d of the segment:

$$\alpha^{\text{med}} = \text{med}_{d \in \text{seg}} \bar{r}_{\ell(d)} \quad (\text{the level of the state holding most days}), \quad (25)$$

$$\alpha^{\text{mean}} = \frac{1}{|\text{seg}|} \sum_{d \in \text{seg}} \bar{r}_{\ell(d)} \quad (\text{day-weighted average of the state levels}). \quad (26)$$

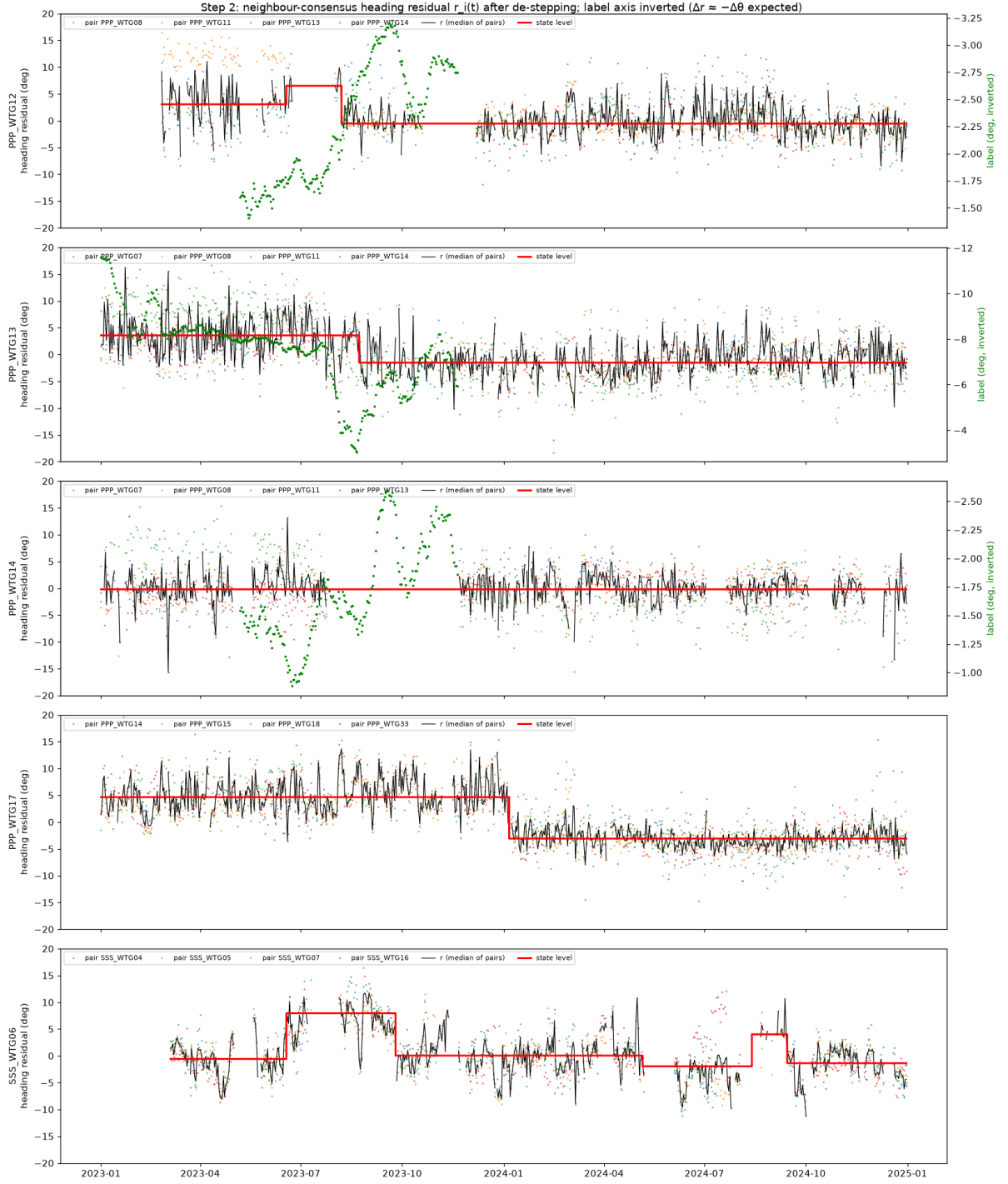


Figure 2: Daily pair residuals $e_{ij}(d)$ (coloured dots), their median $r_i(d)$ (black) and the state levels $\bar{r}_\ell$ (red). On the training turbines the label is drawn in green on an inverted axis, so that r and the label should move together by (13).

The median anchor scores better on train (RMSE 1.91 versus 2.18) because the training turbines have one dominant state. For PPP_WTG17, whose two states hold 370 and 361 days, the median anchor is a coin flip that moves the whole prediction by 4° ; the mean anchor minimises the expected squared error under ignorance of which state is typical, and it was submitted first. Re-anchoring per segment is what stops re-references below the 25° attribution threshold of §4.2 from propagating into the levels.

5.4 Sign checks

Two independent tests are asserted before any file is written; the script aborts if either fails.

1. The least-squares slope of the label on x over the scored training days must be negative; measured -1.07 . This is (13) seen from the labels.
2. On PPP_WTG13, comparing operating samples before 13 January and after 19 January 2023, the median vane moved by -2.6° while the label moved by $+3.0^\circ$. By (11) a change of the set-point (or of δ_v with the controller re-nulling) moves θ by the same amount in the *same* direction, unless the label sign is opposite to the raw vane convention. The observed opposite motion therefore fixes the label as $-v$ in the raw convention, equivalently as $\gamma = W_t - N_t$ once the direction in which this controller reports the vane is accounted for. The test is asserted as “vane and label move in opposite directions” and requires $|\Delta v| > 1^\circ$ and $\Delta\theta^* > 1^\circ$.

Together the two tests pin the sign of the submission model, which is all that matters for scoring.

5.5 Use of labels

No quantity in (23) is fitted to labels: the slope is -1 from (13), the level is (24) and the states come from the SCADA residual alone. The training labels enter in two places only: the sign checks of §5.4, which fix the sign convention of θ , and the evaluation of §6. The segmentation thresholds (λ , $n_{\min}$, the 15° and 25° limits, the 45-day excursion length and the 10° / 91-day transient rule) were set while the training turbines, with their labels, were in view. The frame plan and the transient rule were introduced after the validate round, when the first final file was found to disagree with the raw pair offsets of SSS_WTG06; the files of other participants were used to notice the disagreement and to check the result, not to set a threshold. In particular $\lambda = 400$ was kept although $\lambda = 800$ gives three states on SSS_WTG06, because it is worse on the training turbines (RMSE 2.58 against 2.18).

5.6 What was submitted

file	turbine	states	predicted levels
Results_33_T0_0.csv	PPP_WTG17	2 (boundary 2024-01-06, $\Delta r = -7.74^\circ$)	-6.6° then $+1.1^\circ$
Results_33_T0_final.csv	SSS_WTG06	6, in 1 encoder-free segment	$-8.9^\circ \dots + 1.0^\circ$, mean -1.6°

The six levels on SSS_WTG06 are -0.4 (to 2023-06-17), -8.9 (to 2023-09-24), -1.0 (to 2024-05-05), $+1.0$ (to 2024-08-12), -5.0 (to 2024-09-13) and $+0.4^\circ$. The PPP_WTG17 file of the validate round was produced by the first version of the method (-6.8 then $+1.3^\circ$); the present version changes it by at most 0.16° . Figure 3 shows the predictions against the residual they are derived from.

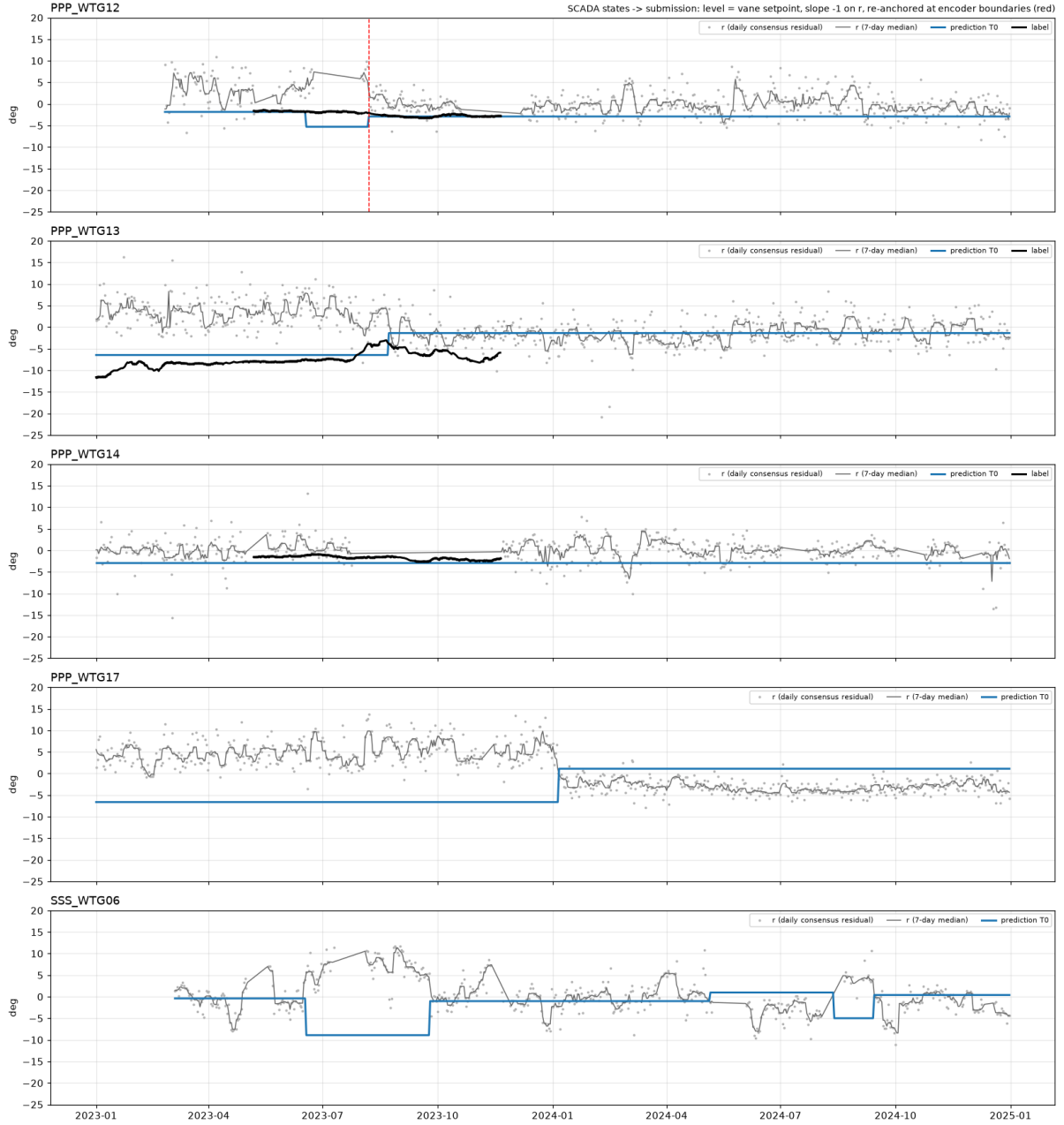


Figure 3: Daily residual r (grey), T0 prediction $\hat{\theta}$ (blue) and, on the training turbines, the label (black). Red dashed lines are encoder-typed boundaries, across which the prediction is re-anchored.

6 Scoring rule and what it rewards

6.1 Metrics

With $\mathcal{D}$ the set of scored days (§2.5),

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} (\hat{\theta}_d - \theta_d^*)^2}, \quad \text{MAE} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} |\hat{\theta}_d - \theta_d^*|, \quad \text{bias} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} (\hat{\theta}_d - \theta_d^*). \quad (27)$$

The tie-breaker is the adjusted Rand index between the submitted clusters and the label-derived states, averaged over turbines. For a contingency table n_{uw} (cluster u , true state w) with margins $n_{u\cdot}$, $n_{\cdot w}$ and $n = |\mathcal{D}|$, Write $I = \sum_{uw} \binom{n_{uw}}{2}$ for the number of day pairs that share both a cluster and a true state, $A = \sum_u \binom{n_{u\cdot}}{2}$ and $B = \sum_w \binom{n_{\cdot w}}{2}$ for the pairs sharing a cluster and sharing a state respectively, and $E = AB / \binom{n}{2}$ for the value of I expected from a random partition with the same margins. Then

$$\text{ARI} = \frac{I - E}{\frac{1}{2}(A + B) - E}, \quad (28)$$

which is 1 for identical partitions up to relabelling and 0 in expectation for a random partition. It is invariant to cluster names; splitting a true state costs ARI in proportion to the split.

6.2 Effective sample size and the block bootstrap

Scored days are runs of a piecewise-constant series, and the error of a piecewise-constant prediction is constant within each run, so the effective number of independent observations is the number of states (two on PPP_WTG17, of order fifteen on the training set), not the number of days. The organisers resample whole states with replacement, score every submission on the same draw, and call a submission better than another only if its RMSE is lower on essentially all draws. The local harness reproduces this with 1000 draws over (turbine, state) blocks; the resulting 95 % interval for the T0 model on the training turbines is 1.5–3.1° around 2.18, which is the honest uncertainty of every score quoted in this note.

6.3 Implication for the strategy

RMSE differences smaller than that band are ties, and ARI then decides. The level of θ cannot be pinned better than a few degrees without labels, so the winnable quantity is the segmentation (ARI, from §4) plus a level that is not badly wrong. The T0 model scores 2.2° on the training turbines against a best-constant baseline of 3.1–6.1°.

7 Summary of assumptions

	measures	units	needs	dominant systematic
r	$\Delta\theta$ from Δr	physical, slope –1	stable healthy neighbours	δ_n , re-references < 15°; neighbour excursions < 10°
L	vane set-point s	physical if $\delta_v = 0$	—	δ_v (WTG13: 2.3°)

8 Outlook: fleet use and further work

8.1 What the method is, operationally

The method needs the nacelle heading and the power of a turbine and of a few neighbours from the standard SCADA stream, no labels and no additional sensor, and processes 16 turbines over

two years in about two minutes. On both held-out turbines it found the large misalignment changes ($8\text{--}9^\circ$) to within a day or two, with the step size in physical degrees. That makes it a candidate for *fleet screening*: a statement of the form “the alignment of this turbine changed by about X° around this date”, which triggers an inspection. It is not a replacement for a measurement campaign, for the reasons below.

8.2 Weaknesses for a fleet roll-out

1. **No absolute level.** By Proposition 1 the residual identifies changes only. The level is the prior (24), which is off by 2.3° on PPP_WTG13 and cannot be checked without a label. A turbine that has been misaligned by a constant amount since commissioning is invisible. This is the main gap: closing it with one calibration per turbine is exactly the cost the label-free tier is meant to avoid.
2. **Noise floor of several degrees.** The day-to-day noise of r is $1\text{--}2^\circ$, but r also wanders by up to $\pm 5^\circ$ over weeks (Figure 3), and with $\lambda = 400$ a change needs $nJ^2/2 \gtrsim 400$ to be accepted. Changes below $3\text{--}5^\circ$ are not reliable. The training labels only contain changes of $1\text{--}3^\circ$, so the range in which the method works is validated on two events (PPP_WTG17, SSS_WTG06) for which no label is available to us.
3. **Dependence on healthy neighbours.** The consensus is a median over four pairs. On SSS_WTG06 two of the four neighbours had excursions of their own in August and September 2024, and the file contains a 32-day state at -5.0° there that is probably not real. Small sites, turbines at the edge of a farm, curtailed or stopped neighbours and neighbours that are themselves being re-aligned are weak spots; an isolated turbine cannot be assessed at all.
4. **Encoder events cost data and can hide a change.** SSS_WTG06 loses 117 of 731 days to excursions and has a residual on 519 days. A misalignment change that happens during an excursion is dated to its end, and one of 15° or more across an excursion is removed as a re-reference. A service visit that changes both the vane and the encoder, the most likely real-world cause of a misalignment change, is exactly the case in which the two are confounded.
5. **Short states are removed.** The transient rule deletes anything shorter than about 45 days that departs by more than 10° and returns. A real fault of that shape, for example a loose vane that is fixed after a month, is not reported.
6. **Hand-set thresholds.** The 25° , 15° , 10° , 45-day, 91-day and λ values were set on five turbines of one type at two sites. The first version failed on the test site for precisely this reason: the validate site did not exercise the de-stepping. Other controllers, encoder types and terrain have to be expected to break some of them.
7. **Sector correction assumes a stationary site.** The sector medians are taken over the whole record. Seasonal stability, vegetation, new neighbouring turbines or a changed curtailment strategy move the heading difference between two turbines without any misalignment.
8. **Retrospective.** Centred rolling medians of 7 and 91 days and a minimum state length of 14 days mean that a change is confirmed about one to two months after it happened. An online version would need one-sided windows and would be noisier.
9. **Changes of the vane gain are invisible.** A change of θ through g in (11) does not move the residual (§4.1).

8.3 Further work

1. **A level estimate.** Combine the residual with a within-turbine, encoder-free estimator of the level, for example the power response during yaw manoeuvres. Such an estimator is noisy day by day but only one number per state is needed: the states of §4.5 say which days may be pooled, which is the averaging the power methods lack. Alternatively, quantify how many spot calibrations per farm are needed when the residual carries the level from one turbine-state to the next.
2. **Farm-wide solution instead of per-turbine medians.** Every pair difference contains the misalignment and the encoder offset of both turbines. Solving all pair series of a site jointly, with one piecewise-constant offset per turbine and robust weights, would use every pair twice, make the attribution of a shift to a turbine explicit and remove the fixed choice of four neighbours and the manual reference blacklist.
3. **Segmentation on valid days with a noise model.** PELT currently runs on a series interpolated across masked days, so that a gap counts as evidence, and with a fixed penalty. Segmenting the valid days only, with a penalty scaled by the turbine’s own residual noise and its autocorrelation, would remove the weak states of 2024 on SSS_WTG06 for a stated reason rather than by tuning λ .
4. **Uncertainty per state.** Report an interval for every $\Delta\theta$ from the spread across pairs and across days, and a flag when a boundary coincides with an excursion of the target or of a neighbour. A screening tool needs to say how much to trust each alarm.
5. **Use the encoder events.** Excursion and re-reference dates are a by-product of the frame plan and are themselves maintenance-relevant; they also mark the days on which a real change is most likely and least observable.
6. **Validation on more labelled turbines.** The claims above rest on three labelled turbines with small steps and two unlabelled ones with large steps. Changes of 3–10° with labels, a second turbine type and a site with fewer than four neighbours are the missing cases.
7. **Online operation.** One-sided windows, a detection delay stated as a function of the step size, and a false-alarm rate estimated on turbines known to be stable.